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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 0.8380989065484519, median 0.7429466314425857, std: 0.5444933620579354
Gyroscope error (imu0):        mean 0.6669782521903689, median 0.6009512231276177, std: 0.361748323013249
Accelerometer error (imu0):    mean 0.48300433415595495, median 0.4296446873985123, std: 0.26624699094655446

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.8380989065484519, median 0.7429466314425857, std: 0.5444933620579354
Gyroscope error (imu0) [rad/s]:     mean 0.004603068670255688, median 0.004147391220697611, std:
0.0024965617195334784
Accelerometer error (imu0) [m/s^2]: mean 0.07390239193330715, median 0.06573806451586812, std:
0.04073729381824298

Transformation (cam0):
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T_ci:  (imu0 to cam0): 
[[ 0.00672062  0.99405019  0.10871547  0.01788196]
 [ 0.9998145  -0.00471729 -0.01867394  0.02002937]
 [-0.01804999  0.10882081 -0.99389749  0.01833657]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0): 
[[ 0.00672062  0.9998145  -0.01804999 -0.01981486]
 [ 0.99405019 -0.00471729  0.10882081 -0.01967648]
 [ 0.10871547 -0.01867394 -0.99389749  0.01665466]
 [ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.029780029168670095


Gravity vector in target coords: [m/s^2]
[-0.65914043 -9.7825262   0.19010003]
```

Calibration configuration
=========================

cam0

Camera model: pinhole
Focal length: [162.92065384228573, 165.6650911402295]
Principal point: [327.59845663186246, 246.0810134801043]
Distortion model: radtan
Distortion coefficients: [-0.24115227477855275, 0.035926627395759365, 0.0024496163453020755, -0.002372523483294585]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.083 [m]
Spacing 0.024900000000000002 [m]

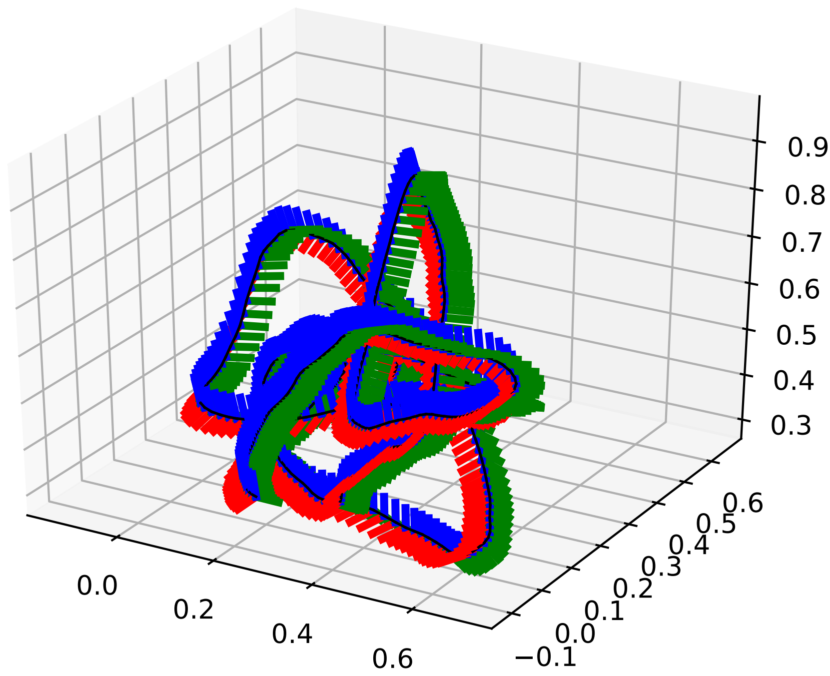
IMU configuration
=================

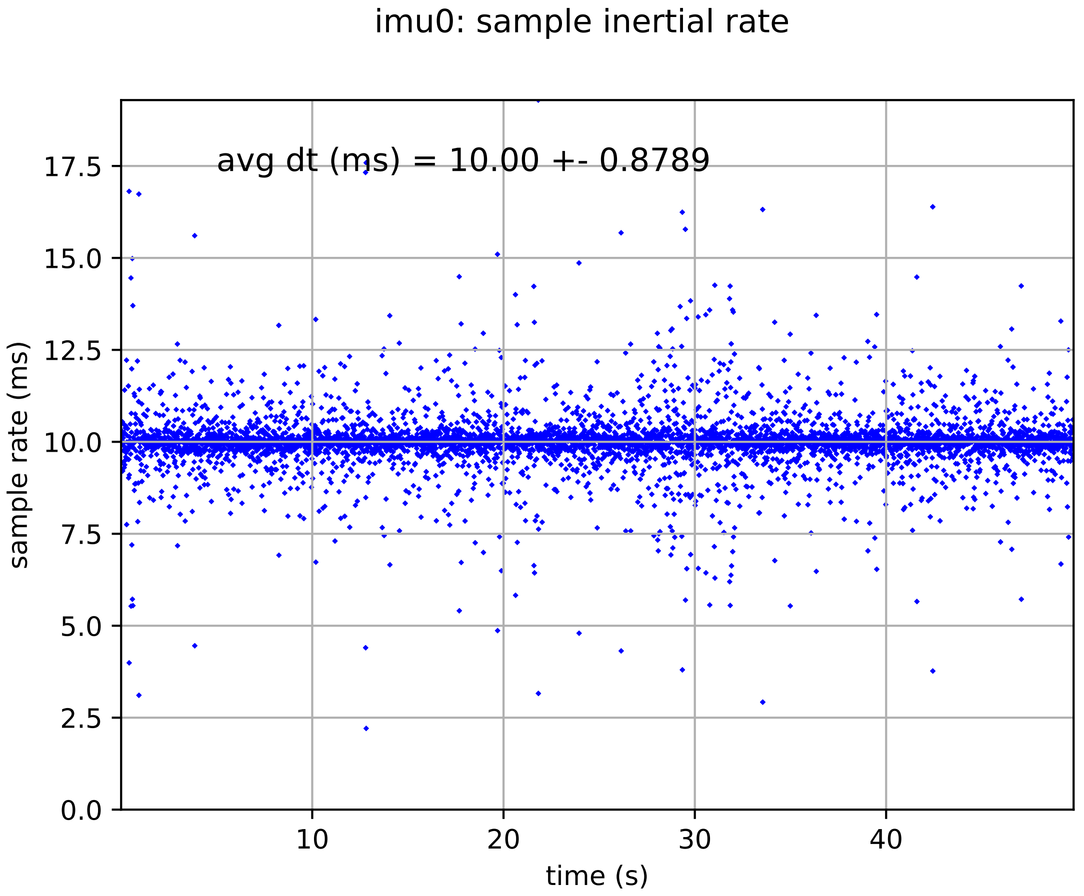
IMU0:

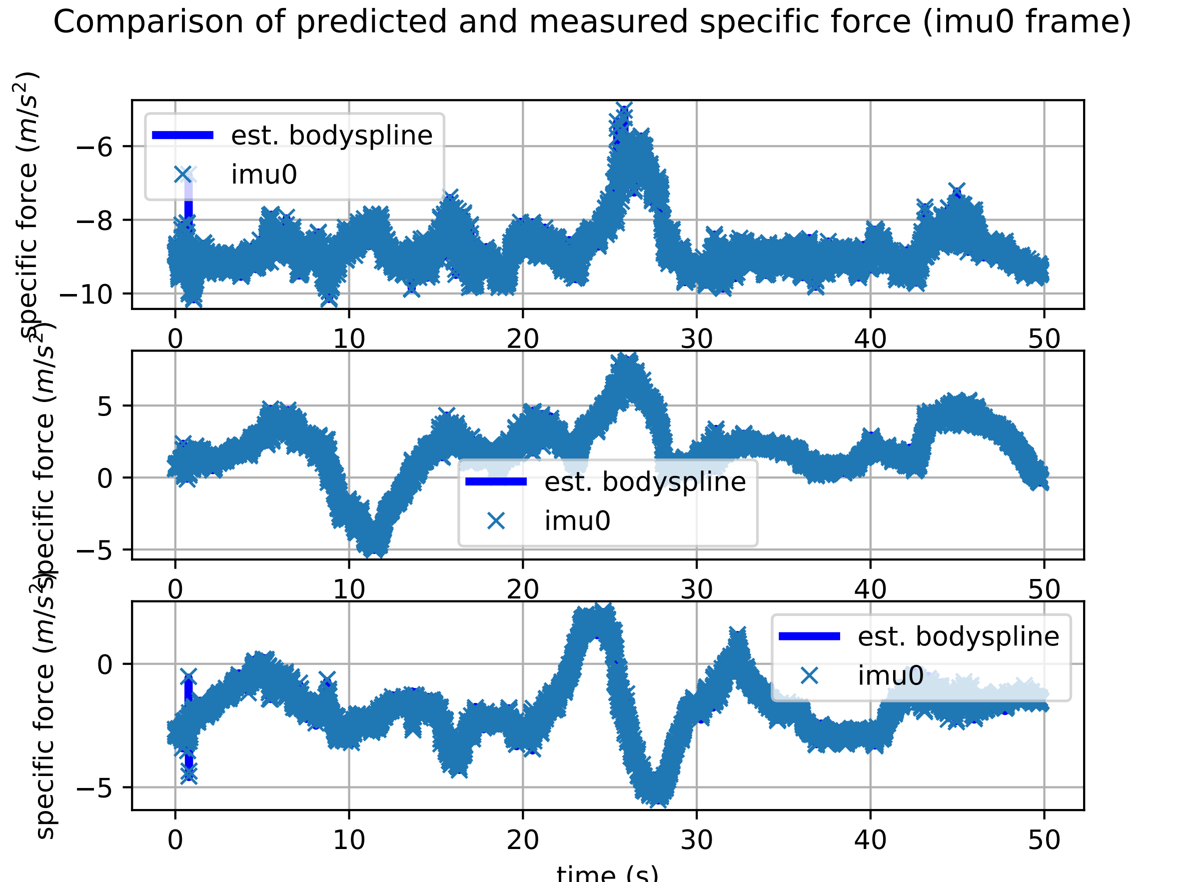
Model: scale-misalignment
Update rate: 100.0
Accelerometer:
Noise density: 0.015300564965416059
Noise density (discrete): 0.15300564965416058
Random walk: 0.001091589605786762
Gyroscope:
Noise density: 0.000690137745142233
Noise density (discrete): 0.00690137745142233
Random walk: 3.678303605230704e-05

```
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
Gyroscope:
M:
[[ 1.01317653  0.          0.        ]
[ 0.05983898  0.97272531  0.        ]
[-0.06487643  0.02861188  1.00792601]]
A [(rad/s)/(m/s^2)]:
[[-0.00189751 -0.00037041 -0.00082622]
[-0.00082313 -0.00154591  0.00217934]
[ 0.00388819  0.00002342  0.00088724]]
C_gyro_i:
[[ 0.99722023  0.06642563 -0.0337558 ]
[-0.06506049  0.99707736  0.04004826]
[ 0.03631737 -0.03774077  0.9986274 ]]
Accelerometer:
M:
[[ 1.01632581  0.          0.        ]
[-0.05195708  1.02664565  0.        ]
[-0.03402404 -0.02818645  0.98770287]]
```

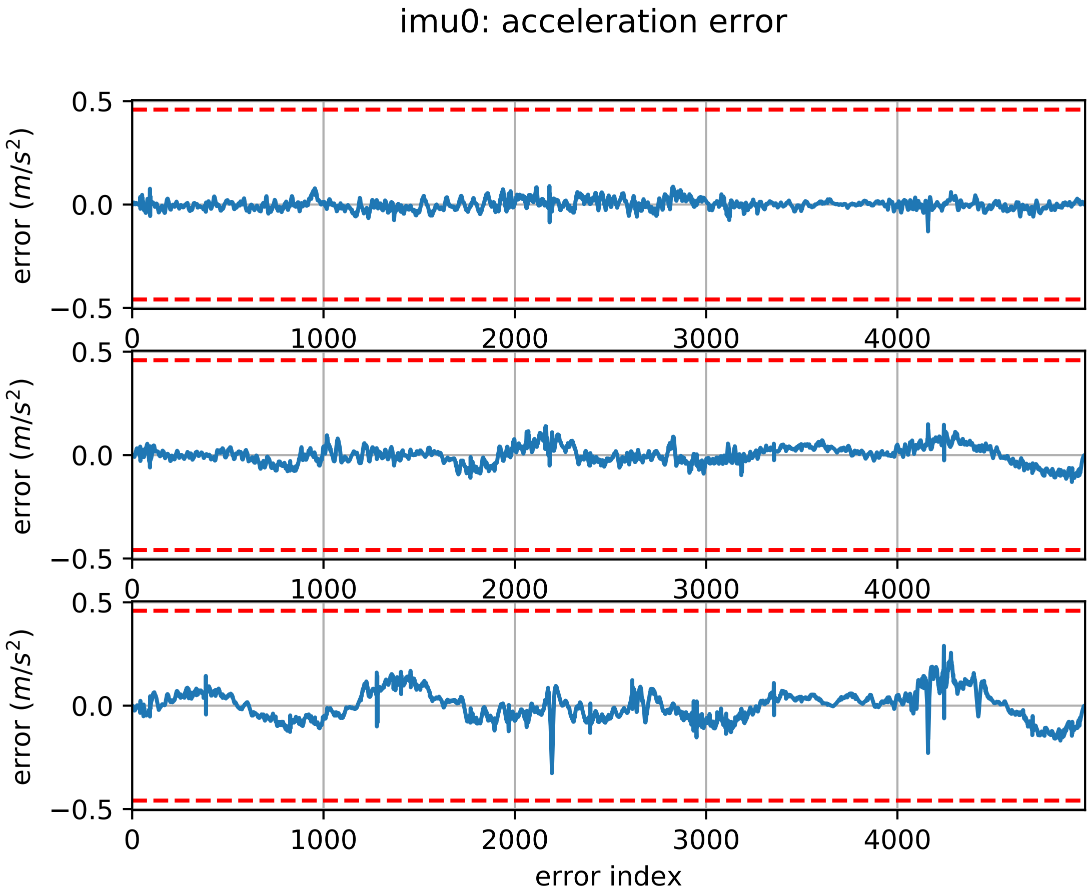
imu0: estimated poses



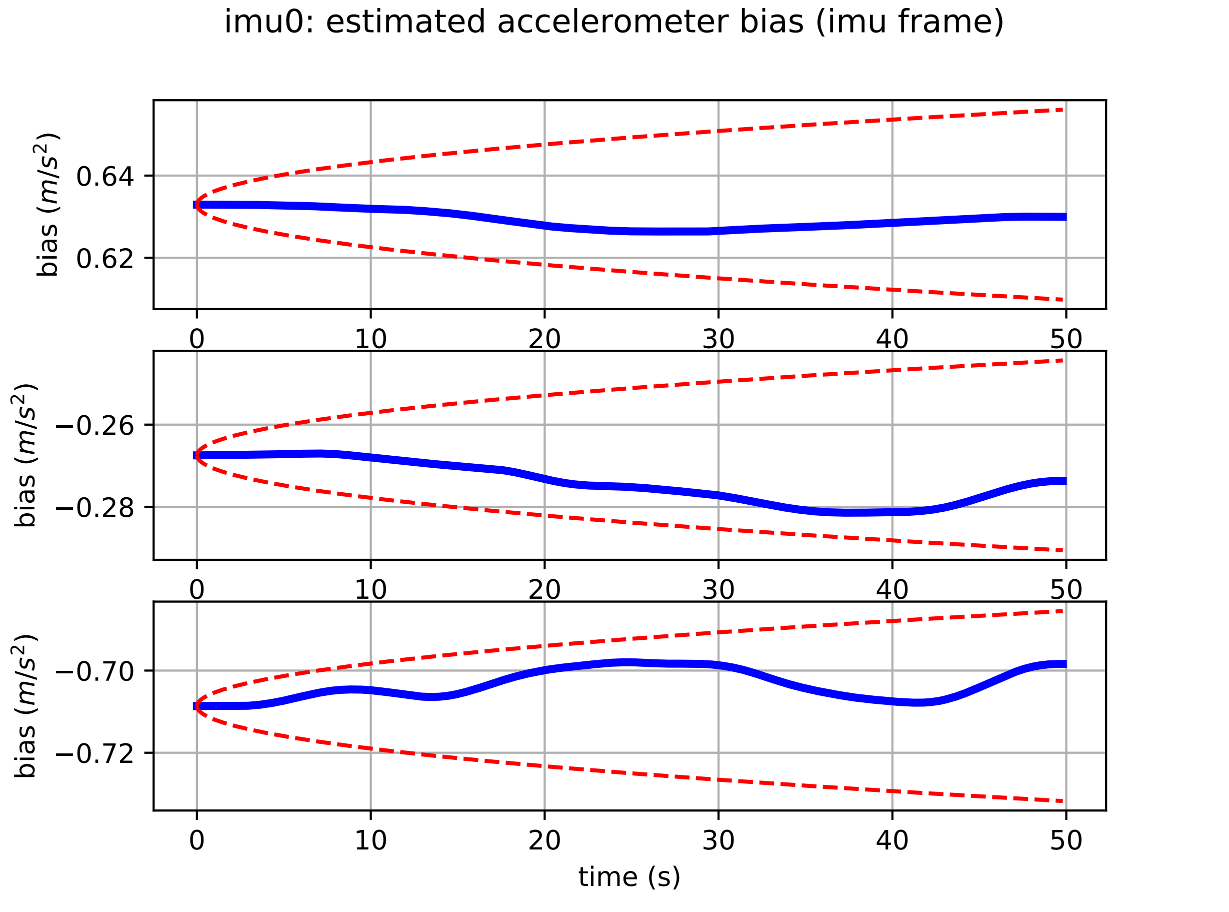

imu0: sample inertial rate
avg dt (ms) = 10.00 +- 0.8789
sample rate (ms)
time (s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
10
20
30
40

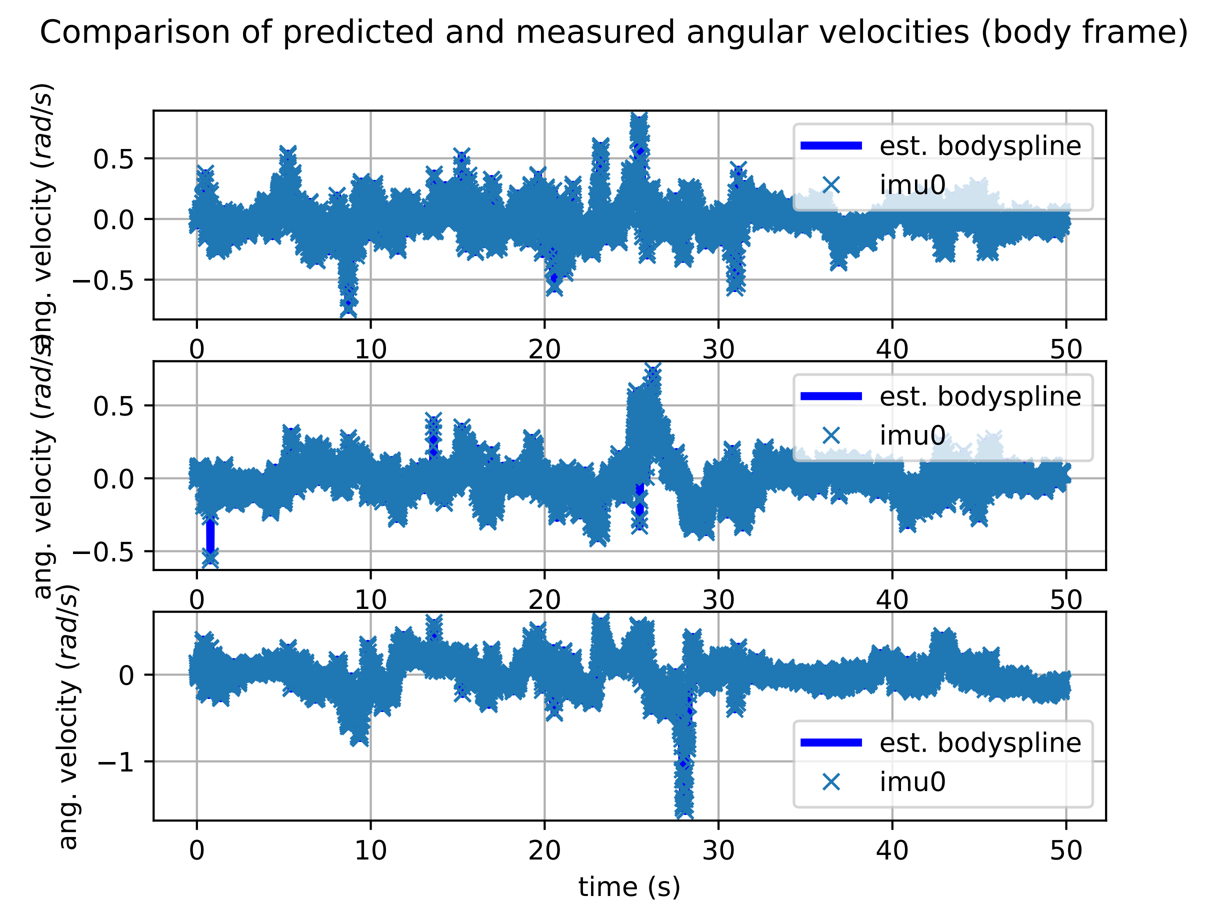


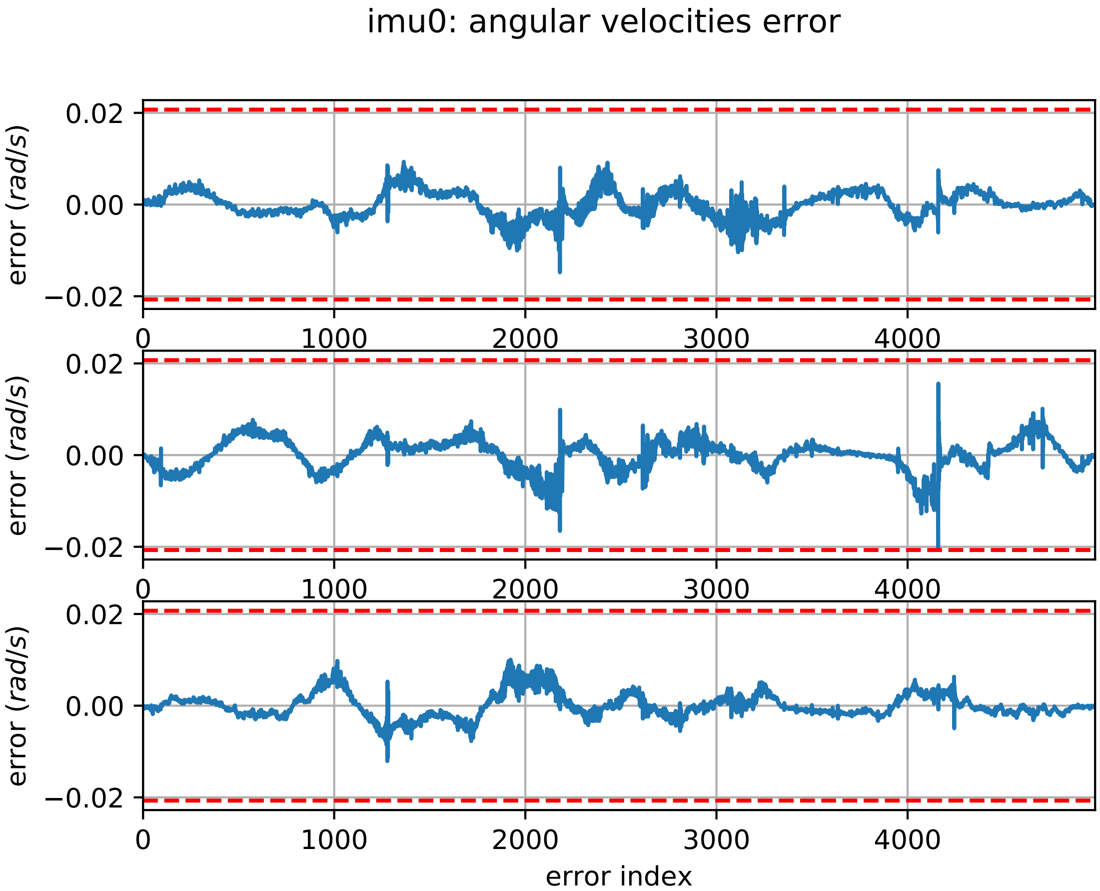
Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
−6
−8
−10
0
10
20
30
40
50
5
0
−5
0
−5
time (s)

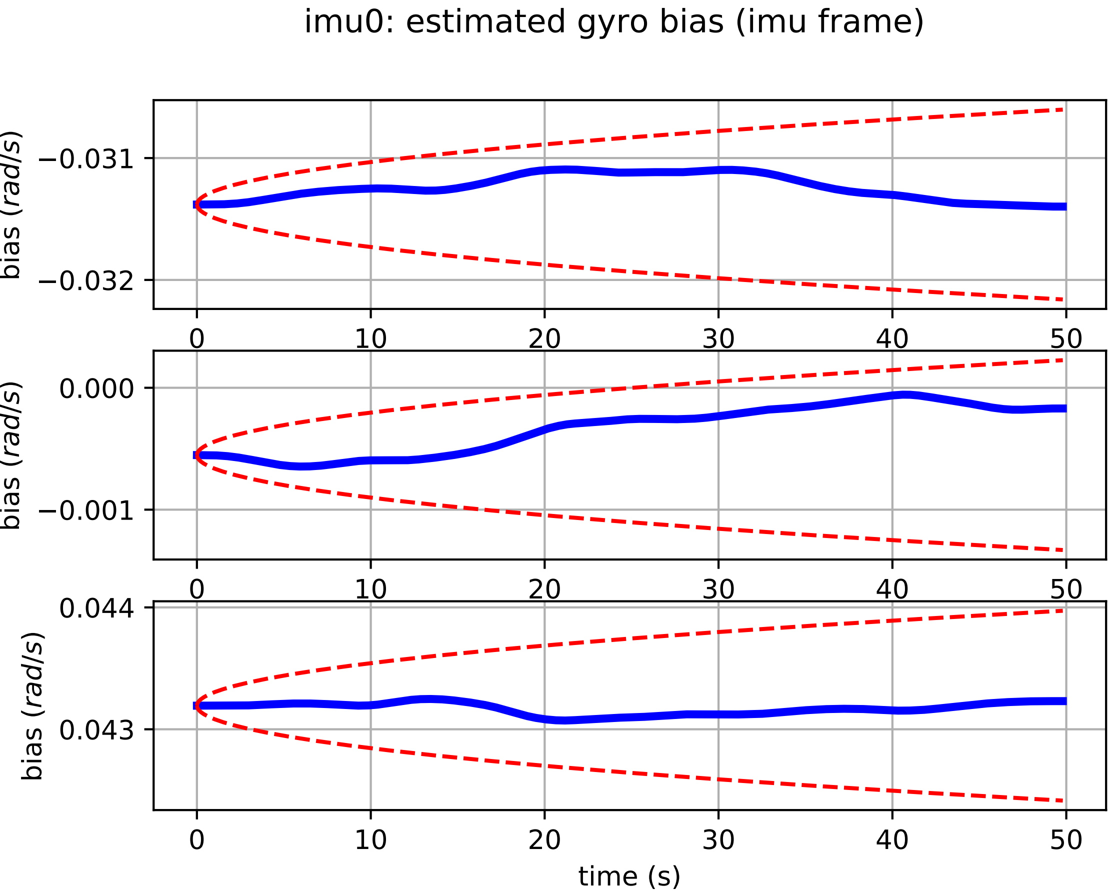


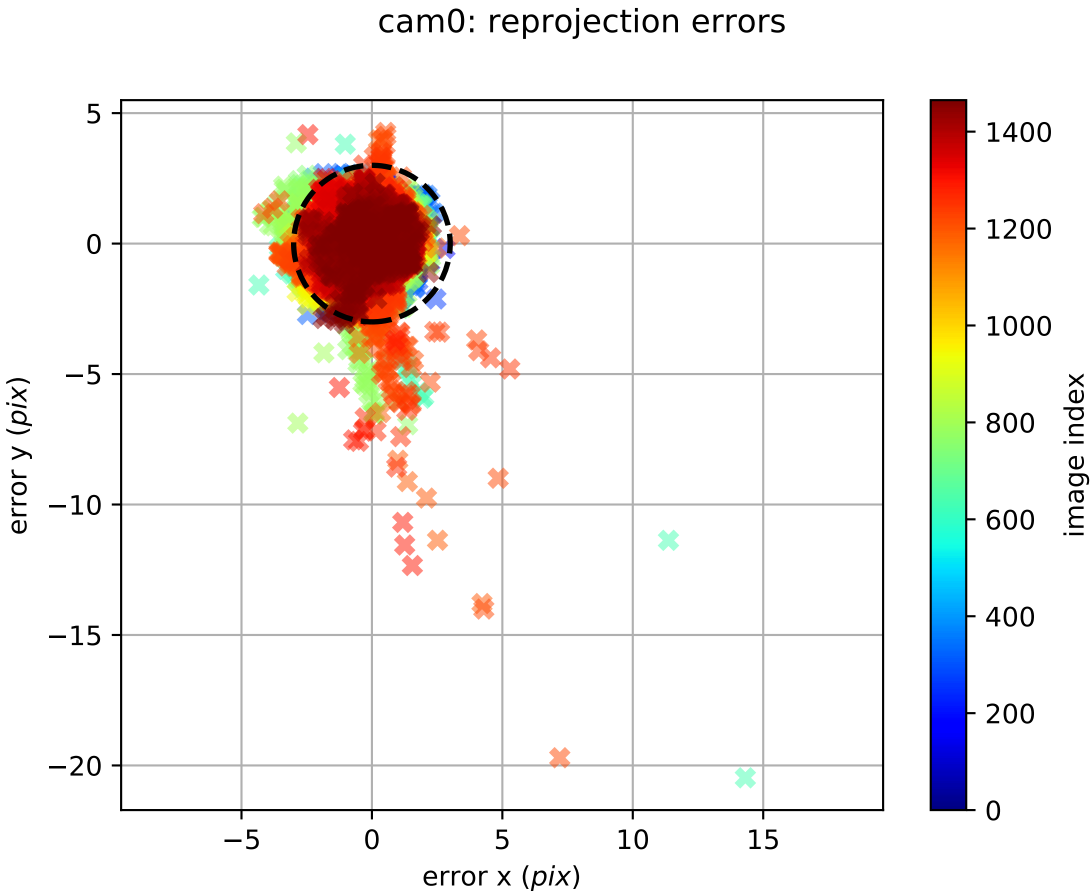
imu0: acceleration error
0.5
0.0
−0.5
error (m/s^2)
0
1000
2000
3000
4000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.64
0.62
−0.26
−0.28
−0.70
−0.72
0
10
20
30
40
50
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
0.5
0.0
−0.5
−1
0
10
20
30
40
50
est. bodyspline
imu0
time (s)


imu0: angular velocities error
error (rad/s)
error index
0.02
0.00
−0.02
0
1000
2000
3000
4000


imu0: estimated gyro bias (imu frame)
bias (rad/s)
−0.031
−0.032
0
10
20
30
40
50
bias (rad/s)
0.000
−0.001
0
10
20
30
40
50
bias (rad/s)
0.044
0.043
0
10
20
30
40
50
time (s)



cam0: reprojection errors
error y (pix)
error x (pix)
image index
5
0
−5
−10
−15
−20
15
10
1400
1200
1000
800
600
400
200